Relative frequency tables

We're familiar now with displaying data in a two-way table. For example, we might categorize the surfboards at Ricky's surf shop by length and by color.

	Blue	White
Longboard	20	15
Shortboard	5	3

But sometimes it's helpful to express the data in a two-way table as percentages. For example, of all the blue surfboards in Ricky's surf shop, what percentage of them are longboards, and what percentage of them are shortboards?

If we want to express these percentages, then we just need to change the two-way table into what's called a **relative frequency table**, which is a table that shows percentages instead of actual counts.

So to answer our question from earlier, we know from our table that there are $20 + 5 = 25$ total blue surfboards, which means that $20/25 = 0.8 = 80\,\%$ of the blue surfboards are longboards. And $5/25 = 0.2 = 20\,\%$ of the blue surfboards are shortboards. We could do the same calculations for the white surfboards:

Longboards: $15/(15 + 3) = 15/18 = 0.8\overline{3} \approx 83\,\%$

Shortboards: $3/(15 + 3) = 3/18 = 0.1\overline{6} \approx 17\,\%$

So we could change the table from one that just shows a count of surfboards, into a relative frequency table that shows the percentage of longboards and shortboards based on color.

	Blue	White
Longboard	0.80	0.83
Shortboard	0.20	0.17
Column total	**1.00**	**1.00**

Now we can see that out of all the blue surfboards, $80\,\%$ of them are longboards and $20\,\%$ of them are shortboards. And out of all the white surfboards, about $83\,\%$ are longboards and $17\,\%$ are shortboards.

We could also create a relative frequency table from one-way table. Using the table from a previous section about the number of times each continent has played host to the summer Olympic games,

Continent	Count
Europe	16
North America	6
Australia	2
Asia	3
South America	1
Column total	**28**

we could turn this table into a relative frequency table.

Continent	Count
Europe	0.57
North America	0.21
Australia	0.07
Asia	0.11
South America	0.04
Column total	**1.00**

Rows vs. columns

Notice that because we found the percent of the total within each column, that we were able to add a "Column total" to the bottom of the surfboard table. Each column sums to 1.00.

$$0.80 + 0.20 = 1.00$$

$$0.83 + 0.17 = 1.00$$

That's because in each cell, we found the percentage of the total for the column, and so each column sums up to $100\,\%$. Therefore, the table we made is actually called a **column-relative frequency**, where the columns sum to 1.00 but the rows do not.

But we could have just as easily created a **row-relative frequency** by converting the rows into percentages instead of the columns into percentages.

We'll start again with the original table.

	Blue	White
Longboard	20	15
Shortboard	5	3

This time, we want to find the percentage of each color based on the kind of surfboard. In other words, what percentage of longboards are blue or white, and what percentage of shortboards are blue or white? Let's do the calculations for longboards:

Blue: $20/(20+15) = 20/35 \approx 0.571 \approx 57.1\,\%$

White: $15/(20+15) = 15/35 \approx 0.429 \approx 42.9\,\%$

Now we'll do the calculations for shortboards:

Blue: $5/(5+3) = 5/8 = 0.625 = 62.5\,\%$

White: $3/(5+3) = 3/8 = 0.375 = 37.5\,\%$

Now we can turn the table into a row-relative frequency table.

	Blue	White	Row total
Longboard	0.571	0.429	**1.00**
Shortboard	0.625	0.375	**1.00**

Out of all the longboard surfboards, $57.1\,\%$ of them are blue and $42.9\,\%$ of them are white. And out of all the shortboard surfboards, about $62.5\,\%$ are blue and about $37.5\,\%$ are white.

Total relative frequency

We can also calculate the total-relative frequency for a two-way table by dividing everything by the grand total.

	Blue	White	Total
Longboard	20	15	**35**
Shortboard	5	3	**8**
Total	**25**	**18**	**43**

Since there are 43 total surfboards, we divide each value in the body of the table and each value in the total column and row by 43. Then we can express the values as decimals

	Blue	White	Total
Longboard	0.46	0.35	**0.81**
Shortboard	0.12	0.07	**0.19**
Total	**0.58**	**0.42**	**1.00**

or as percentages.

	Blue	White	Total
Longboard	46%	35%	**81%**
Shortboard	12%	7%	**19%**
Total	**58%**	**42%**	**100%**